

Racing Against Tomorrow's Safeguards

Is the AI race *really* an arms race?

Pavel Kocourek

pkocourek.com

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“Arms race” $\approx$ a prisoner's dilemma — each lab individually wants to race

“AI Is Not an Arms Race”

Katja Grace, TIME (2023)

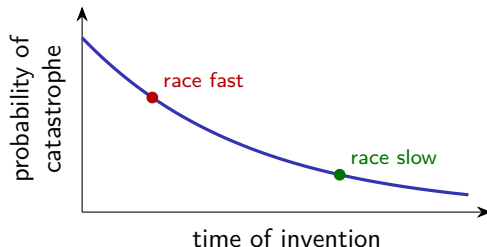
Yet the race is **treated as** one — a **prisoner's dilemma**: everyone races

Does game theory actually support that?

See also *Let's think about slowing down AI* (2022)

Preparedness changes the game

The world grows more prepared:
probability of catastrophe falls over time



Racing faster brings invention into the **high-risk window**;
waiting lets it arrive safer

What's been done | what's new

Prior work

risk held fixed

build / don't-build

Grace 2022

Hendrycks et al. 2023

safety vs. speed

Armstrong et al. 2016

Jensen et al. 2023

growth vs. risk

Jones 2024

dynamic timing, risk constant

Tan 2025

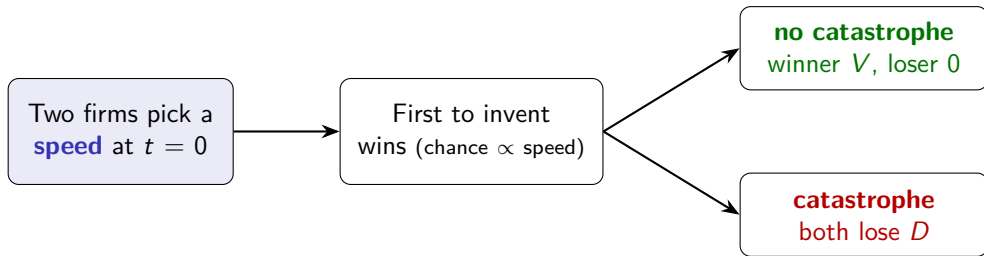
This paper

risk **falls over time**

The simplest dynamic race where catastrophe risk declines as the world grows prepared

→ voluntary slowdown is an **equilibrium**, not unilateral restraint

The model



For a fixed target AI capability, the world keeps getting more prepared — so catastrophe risk **falls over time**

Loury's (1979) patent race — with a chance of catastrophe added

Solution concept — Nash equilibrium

A pair of development speeds where
neither firm can do better
by changing only its own

Result: when risk starts certain

The simple case: effort is free, and catastrophe is certain at the start

They still hold back **voluntarily**

Two observations

- They compete every dollar away — **zero expected payoff**
- Faster preparedness only speeds up the race — catastrophe risk stays the same

Result: the realistic case

The realistic case: catastrophe is likely, but not certain, at the start

Now the game has **two equilibria**

Everyone races

stable: racing is
each firm's best response

Everyone slows

stable too — and
better for everyone

Either can take hold. The binding problem is **coordination**

Calling the AI race an arms race
is a **self-fulfilling prophecy**

The calmer equilibrium is real — the task is to
coordinate on it; even a modest speed limit can tip us there